Confluent Kafka Isotope

This project renders your Kafka pipeline as a **bipartite graph** — services on one vertex set, topics on the other, edges crossing in both directions — so every produce edge, every consume edge, and every terminal consumer is a first-class node in the same topology view. The mechanism is a tracer (an *isotope*) carried in record headers, captured by two library calls: a Kafka **producer interceptor** stamps an isotope and appends one hop per `send()` (produce edges → `hops[]`), and `IsotopeContext.recordConsume(record, service, producer)` forwards that isotope to a value-less marker on `platform.observability.consume_events` (consume edges). Consume-then-produce services additionally call `IsotopeContext.adoptFromRecord(record)` between consume and produce so the trace ID survives each hop. Apache Flink reads only the headers and reports on what the isotopes reveal: **end-to-end latency**, the **full bipartite service⇔topic⇔service graph**, **drop/duplication rates**, and **pipeline coverage**.

A **portability requirement** runs through this project: the same isotope mechanism must work against both **Confluent Cloud for Apache Flink (CCAF)** (managed) and **Confluent Platform for Apache Flink** (self-managed) — both used here via Table API SQL plus uploaded UDF/PTF JARs (no DataStream code on either side). Three decisions follow from that: tagging happens in a Kafka **producer interceptor** (the one extension point both runtimes share via the broker), the on-wire **header** format is **JSON** (so Flink SQL can read the scalar fields with `CAST(headers[…] AS STRING)` and no UDF), and the optional stateful reports (`LatencyPercentilesUDAF`, `StuckTracePTF`) ship as a single JAR that registers identically on either runtime.

One asymmetry the runtimes don't share: **Flink-native SR-Protobuf**. CCAF supports SR-framed Protobuf as a Flink sink format via its topic catalog; Apache Flink open-source (the CP Flink runtime) ships `avro-confluent` but no SR-Protobuf counterpart. So Flink *report sinks* land on **Avro+SR on CP** and can be **Protobuf+SR on CCAF** — a runtime constraint, not a project preference. The demo *event* topics (next paragraph) are unaffected because they're written by the Kafka producer client, not by Flink.

Message **values** on the demo topics are **SR-framed Protobuf** (`ai.signalroom.kafka.isotope.proto.DemoEvent`) — the standard Confluent value format. The interceptors and reports are agnostic to value format because the isotope rides in headers; the Protobuf choice just gives the integration tests and the demo CLI a typed payload to work with.

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1.0 How the isotope is carried

- **Header `x-isotope`** (JSON bytes) carries the full hop history, forwarded by every hop:

- `t` — 16-byte **UUIDv7** trace ID (RFC 9562): 48-bit ms timestamp in the high bits + 74 bits random. Stable for the life of the trace,and lexicographic byte order matches creation order — sort trace IDs and you get chronological order for free.
- `o` — origin timestamp (ms) — same value as the timestamp embedded in the UUIDv7 trace ID; kept as its own field for typed access from Flink SQL without needing to decode the UUID bytes.
- `s` — origin service name (set once, never reassigned)
- `h` — ordered list of hops, each `{s: service, t: topic, m: tsMs}`
- `x` — `true` if the hop list exceeded `MAX_HOPS = 32` and the oldest hop was evicted

- **Six scalar headers** (UTF-8 strings) carry the most-recent-hop view so Flink SQL can read them via `CAST(headers['x-isotope-…'] AS STRING)` without parsing the JSON array (no UDF required on either CCAF or CP Flink). See scripts/flink/README.md for the full header table.

A producer with the isotope interceptor loaded appends one hop on every `send()`. A consume-then-produce service calls `IsotopeContext.adoptFromRecord(record)` between consume and produce so the trace ID and origin survive the hop.

1.1 How the producer interceptor gets invoked

How the producer interceptor gets invoked. Application code never calls `onSend` / `onAcknowledgement` directly — the Kafka client invokes them by reflection once two things are in place:

1. **Register the class in producer config.** Put `IsotopeProducerInterceptor.class.getName()` under `interceptor.classes` (App.java:199, App.java:284, IsotopeTestHarness.java:96). The `KafkaProducer` constructor instantiates the interceptor and owns its lifecycle.
2. **Call `producer.send(...)` as normal.** Every `send()` triggers `onSend` (caller thread, before serialization) and later `onAcknowledgement` (producer I/O thread, after broker ack or failure).

The exact call sites in `kafka-clients` 4.2.0: `KafkaProducer.send` invokes `interceptors.onSend` (line 950) and `AppendCallbacks.onCompletion` invokes `interceptors.onAcknowledgement` (line 1600). Exceptions thrown by the interceptor are caught and logged by the client's fan-out wrapper — they never break the `send` call.

1.2 Using the Kafka client interceptor vs explicit library calls

This project uses one Kafka client interceptor, and it's a `ProducerInterceptor` — not a `ConsumerInterceptor`. A `ConsumerInterceptor.onConsume` sees a whole batch from `poll()`, but the application processes records one at a time, so a thread-local snapshot from the batch would be ambiguous about which record is being handled. An earlier `IsotopeConsumerInterceptor` was tried and removed for exactly that reason (see #30). The consume side instead runs on **explicit per-record library calls**: services call `IsotopeContext.adoptFromRecord(record)` between consume and produce to carry the trace through a hop (the next `send()` then sees that thread-local and appends a new hop), and `IsotopeContext.recordConsume(record, service, markerProducer)` to emit a consume-edge marker (see the next paragraph). So: **producer interceptor for produce-side stamping; explicit per-record calls for everything consume-side**.

1.3 Why explicit calls for consume-side markers?

Consume-side markers — the other half of the bipartite graph. The produce-side `hops[]` chain captures `producer → topic` edges and implicitly captures consume edges for *intermediate* services (svc-B consuming from topic-AB is implied by svc-B's next produced hop), but **terminal consumers** — services that consume but never produce — would otherwise be invisible. The bipartite story closes that gap with one library call: consumers call `IsotopeContext.recordConsume(record, service, markerProducer)` between consume and process, which forwards the inbound record's six scalar `x-isotope-*` headers to a value-less marker record on `platform.observability.consume_events` and adds one new header — `x-isotope-consumer-service` — that names the consumer. The `bipartite_topology` Flink report unions these markers with the produce-side view to emit edges in both directions: `producer → topic` (from `isotope`) and `topic → consumer` (from `consume_events`).

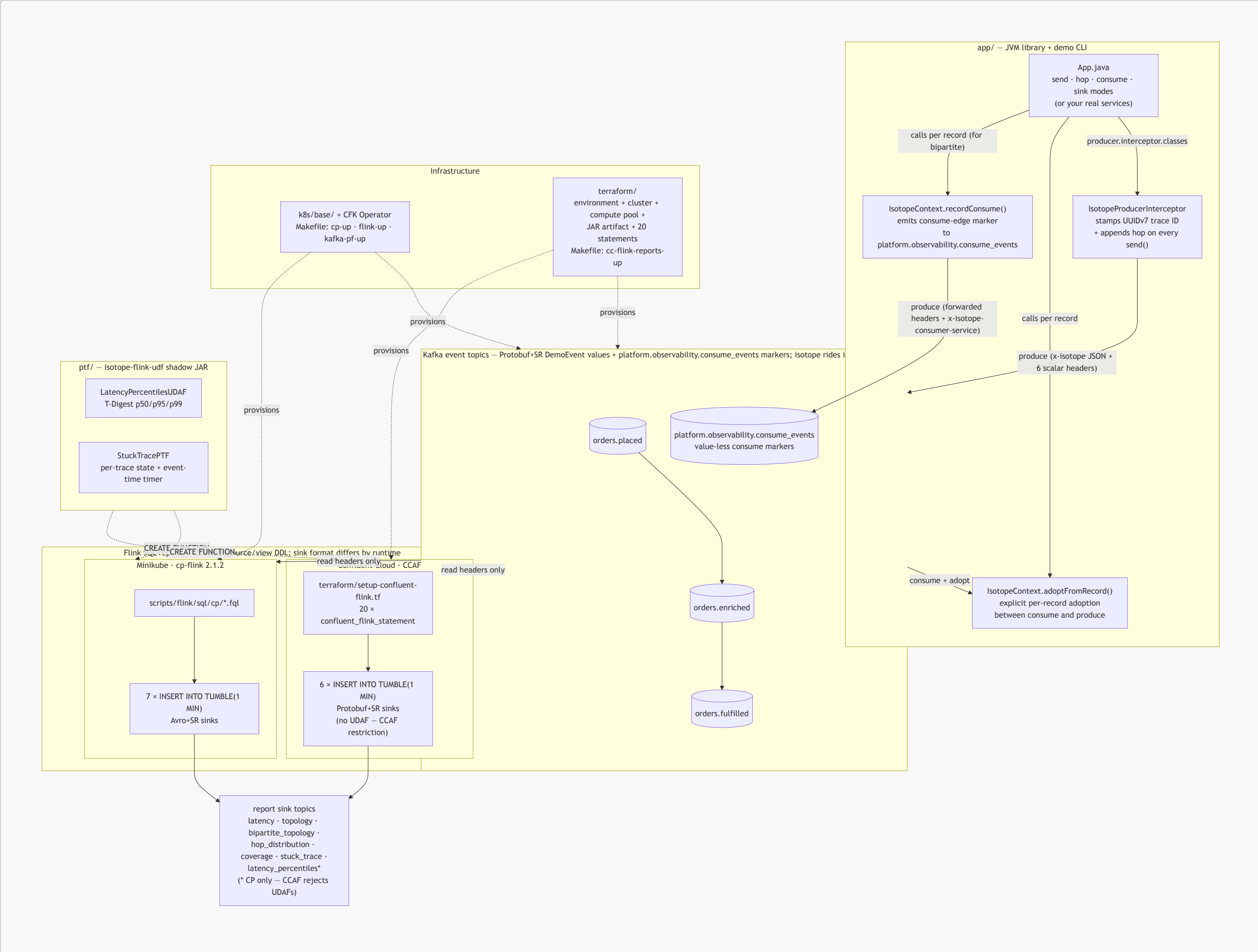
Markers are fire-and-forget: a dropped marker leaves a hole in the topology graph but never disrupts the consume/produce pipeline.

1.4 Why "bipartite"?

Background — why "bipartite"? The resulting topology report is an example of a **bipartite graph** from graph theory (a sub-field of discrete mathematics). A graph is *bipartite* when its vertices partition into two disjoint sets such that every edge connects a vertex in one set to a vertex in the other — edges never run within a set. Here the two sets are **services** (`svc-A`, `svc-B`, …) and **topics** (`orders.placed`, `orders.enriched`, `orders.fulfilled`, `platform.observability.consume_events`); every edge is either a **produce edge** (`service → topic`, from the isotope's `hops[]`) or a **consume edge** (`topic → service`, from the `platform.observability.consume_events` markers). Kafka's pub/sub model guarantees the bipartite shape: services never connect directly to other services, and topics never connect directly to other topics — every interaction flows through the opposite set. Before consume-side markers existed, only produce edges were captured, so the topology view collapsed into a **unipartite** `service → service → service` chain with topics hidden as edge labels and terminal consumers omitted entirely. With both edge directions now wired, topics become first-class nodes alongside services, and the `bipartite_topology` report renders the full graph — every produce *and* consume edge, in both directions.

2.0 Architecture

A bird's-eye view of the moving parts. The JVM library in app/ registers a Kafka producer interceptor that stamps the isotope into record headers on every `send()`; consume-then-produce services adopt the inbound trace via an explicit `IsotopeContext.adoptFromRecord(record)` call; records flow through a 3-topic chain; Flink SQL reads only the headers and emits 1-minute aggregate reports. The same source/view DDL deploys to both runtimes — **CP** on Minikube applies `.fql` files under scripts/flink/sql/cp/, and **CCAF** in Confluent Cloud applies inline `confluent_flink_statement` resources under terraform/. The shadow JAR from ptf/ (which powers two of the six reports) registers identically on both. (Kafka is drawn once below for brevity — each runtime provisions its own cluster.)



See § 3.0 for the file tree behind each box, and § 4.0 for the run commands.

3.0 Repo layout

```
app/                                    isotope JVM library + demo CLI + tests
  src/main/proto/ai/signalroom/kafka/isotope/proto/
    demo_event.proto                    DemoEvent message (Protobuf value schema)
  src/main/java/ai/signalroom/kafka/isotope/
    Isotope.java                        POJO + JSON codec + Hop + fromHeaders()
                                        + UUIDv7 helpers (uuidV7Bytes / uuidV7String)
    IsotopeContext.java                 ThreadLocal + adoptFromRecord() +
                                        recordConsume() (emits consume-edge markers)
    IsotopeProducerInterceptor.java     stamps/appends x-isotope + 6 scalar
                                        reporting headers on send()
    App.java                            demo CLI — send / hop / consume / sink modes
  src/test/java/.../                    IsotopeCodecTest,
IsotopeContextRecordConsumeTest
                                        (no broker needed)
  src/integrationTest/java/.../         BrokerSmokeIT, ProducerInterceptorIT,
                                        ThreeStageHopPropagationIT,
BipartiteTopologyIT,
                                        IsotopeTestHarness — live-broker tests;
produce/consume
                                        DemoEvent via SR-framed Protobuf
                                        (need Minikube CP + SR port-forwarded)
ptf/                                    Flink PTF + UDAF shadow JAR (powers 2 of 6
reports)
  src/main/java/ai/signalroom/kafka/isotope/flink/
```

```
    LatencyPercentilesUDAF.java         T-Digest p50/p95/p99 aggregate
    StuckTracePTF.java                  per-trace state + event-time timer
    Percentiles.java                    p50/p95/p99 return-type DTO
  src/test/java/.../                    LatencyPercentilesUDAFTest
k8s/base/                               CFK manifests
  confluent-platform-c3++.yaml          Kafka / SR / Connect / ksqlDB / Control Center
  flink-basic-deployment.yaml           cp-flink session cluster + CMF
  flink-rbac.yaml                       RBAC for the cp-flink operator
scripts/
  port-forward-kafka.sh                 localhost:30092 → Kafka, localhost:8081 → SR
  port-forward-taskmanager.sh           Flink TaskManager web UI forward
  deploy-cp-flink-reports.sh            builds shadow JAR + applies sql/cp/*.fql to
                                        the cp-flink session cluster
  deploy-cc-flink-reports.sh            builds shadow JAR + wraps `terraform apply`
                                        for the CCAF path
  cc-cli-env.sh                         pulls Kafka + SR creds from `terraform
output`,
                                        builds the JAAS string, exports BOOTSTRAP /
                                        SR_URL / KAFKA_KEY / KAFKA_SECRET / JAAS / ...
  cc-app-run.sh                         thin wrapper around `./gradlew :app:run` that
                                        sources cc-cli-env.sh and injects the six -D
flags
  flink/README.md                       Flink SQL reports — runtime split (CP=7
reports/Avro+SR,
                                        CCAF=6 reports/Protobuf+SR), layout,
operations
  flink/sql/cp/                         CP Flink SQL: 00_source_table,
01_register_functions,
                                        05_isotope_view, 06_consume_events_view,
                                        05_report_sinks (avro-confluent),
                                        10/20/25/30/40/60/70 INSERT INTO reports,
99_teardown
                                        (CCAF SQL is inlined under terraform/setup-
confluent-flink.tf.)
terraform/                              CCAF infrastructure-as-code (`make cc-flink-
reports-up`)
  providers.tf                          Confluent provider — cloud key/secret vars
  versions.tf                           required Terraform (>= 1.13) + provider
versions
  variables.tf                          confluent_api_key/secret, cloud, region,
day_count
  data.tf                               organization lookup + other data sources
  setup-confluent-environment.tf        environment (ESSENTIALS stream-governance
package)
  setup-confluent-kafka.tf              Kafka cluster + Kafka API key rotation module
                                        (iac-confluent-api_key_rotation-tf_module)
  setup-confluent-flink.tf              service account + 6 role bindings, compute
pool,
                                        artifact upload, SR API key rotation, and 20
inline
                                        `confluent_flink_statement` resources: 4 ALTER
TABLE
                                        + 3 VIEW + 6 sink CREATE TABLE + 1 CREATE
FUNCTION
                                        (PTF; UDAF skipped — CCAF rejects UDAFs) + 6
INSERT INTO
  outputs.tf                            environment_id, bootstrap, SR URL, rotating
                                        Kafka + SR API key/secret outputs (sensitive)
  terraform.png                         rendered resource graph (embedded in § 4.5)
Makefile                                cp-up / flink-up / kafka-pf-up / flink-
reports-up /
```

```
                                                  cc-flink-reports-up / cc-flink-reports-down /
...
```

4.0 Running

Cheapest-first order if you're new: `./gradlew test` (§ 4.1) → local CP via Minikube (§ 4.2–4.4) → CCAF in the cloud (§ 4.5). Skip ahead if you only care about one runtime.

4.1. Unit tests (no broker, instant)

```
./gradlew test                         # both subprojects
# or scoped:
./gradlew :app:test                    # IsotopeCodecTest (10) +
IsotopeContextRecordConsumeTest (4) — JSON roundtrip, hop eviction, UUIDv7 properties,
consume-marker emission (14 tests)
./gradlew :ptf:test                    # LatencyPercentilesUDAFTest — T-Digest
accumulator semantics
```

4.2 Demo CLI — see one trace propagate live

The fastest way to watch the isotope mechanic. Requires the cluster to be up and the Kafka + SR forwards running (see step 3 below for the bring-up commands). The CLI has two argument styles — **pipeline-position verbs** that bake in the orders.* topic chain (recommended for the demo) and **generic verbs** that take raw topic + service args (for ad-hoc inspection on any topic):

Verb	Args	What it does
`place`	`[payload]`	Produces one isotope-tagged `DemoEvent` to `orders.placed` as `order-intake-service` (default payload: `hello`), then exits. Auto-creates the topic.
`enrich`	—	Consumes from `orders.placed`, adopts the isotope, **emits a consume-edge marker to `platform.observability.consume_events` as `order-enrichment-service`**, then re-produces to `orders.enriched`. Runs until Ctrl-C.
`fulfill`	—	Same as `enrich` but for `orders.enriched → orders.fulfilled` as `order-fulfillment-service`. Runs until Ctrl-C.
`ship`	—	Terminal consumer for `orders.fulfilled` as `shipping-notification-service`. **Emits a consume-edge marker** so it shows up in the bipartite report; does not re-produce. Runs until Ctrl-C.
`send`	`<topic> <service> <payload>`	Generic produce. Auto-creates the topic.
`hop`	`<in-topic> <out-topic> <service>`	Generic consume-then-produce; emits a consume-edge marker. Runs until Ctrl-C.
`consume`	`<topic> <service>`	Generic terminal-consume; emits a consume-edge marker and pretty-prints the trail. Runs until Ctrl-C.
`sink`	`<topic>`	Passive peek — pretty-prints the isotope trail but does NOT emit a consume marker. Use for ad-hoc inspection. Runs until Ctrl-C.

A 4-stage chain (full bipartite graph) in four terminals:

```
# Terminal A — terminal consumer (prints the full 3-hop trail AND emits a
#              consume-edge marker so shipping-notification-service shows up
#              in the bipartite report)
./gradlew :app:run --args="ship" -q

# Terminal B — middle stage: orders.enriched → orders.fulfilled
./gradlew :app:run --args="fulfill" -q

# Terminal C — first stage: orders.placed → orders.enriched
./gradlew :app:run --args="enrich" -q

# Terminal D — kick the chain off (run repeatedly to send more)
./gradlew :app:run --args="place 'hello world'" -q
```

Terminal A's output for each record shows the same `trace_id` across all three hops, `origin = order-intake-service` (never reassigned), and `hops[]` listing `order-intake-service → order-enrichment-service → order-fulfillment-service` in order with per-hop timestamps. The bipartite-topology report sees all six edges: produce edges `order-intake-service → orders.placed`, `order-enrichment-service → orders.enriched`, `order-fulfillment-service → orders.fulfilled` and consume edges `orders.placed → order-enrichment-service`, `orders.enriched → order-fulfillment-service`, `orders.fulfilled → shipping-notification-service`. Swap `consume` for `sink` if you only want to inspect records without recording the terminal edge. Override endpoints via `-Dkafka.bootstrap=…` / `-Dschema.registry.url=…` if you're not on the default Minikube layout.

4.3 Integration tests (live Kafka via Minikube)

Bring up the local Confluent Platform stack and port-forward Kafka + SR:

```
make minikube-start                  # one-time
make cp-up                           # CFK Operator + Kafka/SR/Connect/ksqlDB/C3 (~5 min)
make kafka-pf-up                     # localhost:30092 → Kafka, localhost:8081 → Schema Registry
```

Then run the suite:

```
./gradlew :app:integrationTest                                  # every IT below
./gradlew :app:integrationTest --tests '*ProducerInterceptorIT'         # just one
```

Override the endpoints if needed:

```
./gradlew :app:integrationTest \
    -PkafkaBootstrap=localhost:30092 \
    -PschemaRegistryUrl=http://localhost:8081
```

Tear down forwards when done:

```
make kafka-pf-down
```

The integration tests cover:

Test	What it verifies
`BrokerSmokeIT`	AdminClient can create/list/delete a topic via the NodePort port-forward
`ProducerInterceptorIT`	A consumer sees the `x-isotope` JSON header + all 6 scalar reporting headers with the expected origin/hop values, and the Protobuf round-trip preserves `DemoEvent.source` / `payload`
`ThreeStageHopPropagationIT`	`order-intake-service → topic-AB → order-enrichment-service → topic-BC → order-fulfillment-service` produces a stable trace ID, 2-hop trail in send order, and correct scalar headers (origin = `order-intake-service`, this = `order-enrichment-service`, hop count = 2) at the terminal; consume-then-produce hops use `IsotopeContext.adoptFromRecord` to carry the trace forward
`BipartiteTopologyIT`	The 4-stage `order-intake-service → topic-AB → order-enrichment-service → topic-BC → order-fulfillment-service → topic-CD → shipping-notification-service` chain emits exactly three consume-edge markers to a per-test markers topic — one per consume edge. Every marker carries the trace ID, forwarded `x-isotope-*` scalars describing the upstream producer, and the new `x-isotope-consumer-service` naming the downstream consumer. Asserts the `(consumer_service, consumed_topic)` set is exactly the three pairs of stages 2-4

4.4 Flink SQL reports on Confluent Platform for Apache Flink (Minikube)

The five pure-SQL reports plus the two JAR-backed reports (one PTF, one UDAF) — seven in total — run against a Flink session cluster managed by the Confluent Flink Kubernetes Operator. Same FQL files deploy to Confluent Cloud for Apache Flink — see **§ 4.5** for that path; this section is the local-Minikube path.

Bring up Flink:

```
make flink-up                # cert-manager → CFK Flink Operator → CMF → session cluster
                             # (~5 min the first time)
```

Deploy the reports — all 7 reports run on the cp-flink session cluster (Flink 2.1.2). Sink topics use Apache Flink's `avro-confluent` format — SR-framed Avro, auto-registered on first write — so Control Center renders the report rows natively.

```
make flink-reports-up
```

`flink-reports-up` builds the PTF/UDAF shadow JAR if missing, copies it into the JobManager pod, pre-creates the 7 sink Kafka topics, then applies the source + view + sink DDL and submits 7 `INSERT INTO` streaming jobs (one per report). On first write to each sink, Apache Flink's `flink-sql-avro-confluent-registry` format registers a fresh Avro schema in SR under subject `<topic>-value`. **Control Center deserializes all 7 report topics natively** — no `.proto` files in the repo for the reports, no hand-installed format jars beyond the one init-container download.

4.4.1 Format-by-domain

The demo *event* topics (`orders.placed`, `orders.enriched`, `orders.fulfilled`) still ride **Protobuf+SR** via the Java app's `DemoEvent` schema — that's unchanged. The consume-edge marker topic `platform.observability.consume_events` has **null value + scalar headers only** (Flink reads the headers via a `MAP<STRING, BYTES>` virtual column; there's nothing to deserialize). The *report* topics ride **Avro+SR** because cp-flink doesn't ship an SR-integrated Protobuf format and CMF (which does) disallows the UDAFs the percentiles report needs. Events from the app are Protobuf; aggregates from Flink are Avro; consume markers are headers-only. Format by domain — a clean split, not a defect.

4.5 Flink SQL reports on Confluent Cloud for Apache Flink (CCAF)

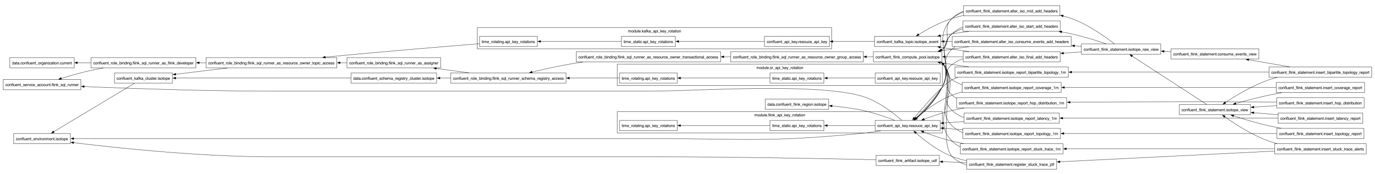
CCAF parallel of § 4.4, driven by Terraform under terraform/. One `make` target spins up a fresh Confluent Cloud environment, Kafka cluster, 4 pre-created event topics (the three demo topics `orders.placed` / `orders.enriched` / `orders.fulfilled` plus the consume-edge marker topic `platform.observability.consume_events`; report sink topics are created on the fly by the Flink `CREATE TABLE` statements — see the comment in terraform/setup-confluent-kafka.tf for why pre-creating them via `confluent_kafka_topic` would conflict with CCAF's Topic Catalog auto-import), a Flink compute pool, two rotating service-account API key pairs (one for Kafka, one for Schema Registry), the PTF/UDAF JAR uploaded as a Flink artifact, and 20 long-lived `confluent_flink_statement` resources broken down as **4 ALTER TABLE** (add scalar headers on the event topics) + **3 VIEW** (raw + typed produce + typed consume) + **6 sink CREATE TABLE** + **1 CREATE FUNCTION** (PTF only — the UDAF is intentionally skipped, see the limitation note below) + **6 INSERT INTO** streaming jobs. The Terraform shape mirrors `apache_flink-kickstarter-ii` — same provider version, same `iac-confluent-api_key_rotation-tf_module`, same DROP-then-CREATE statement pattern.

Prereqs:

- Terraform `>= 1.13` installed locally.
- A Confluent Cloud API key (Cloud-level, not cluster-scoped) with permissions to create environments, Kafka clusters, Flink compute pools, service accounts, role bindings, Flink artifacts, and statements. Generate via Console → Settings → Cloud API keys.

Deploy:

```
export CONFLUENT_API_KEY=...
export CONFLUENT_API_SECRET=...
make cc-flink-reports-up CONFLUENT_API_KEY=$CONFLUENT_API_KEY
CONFLUENT_API_SECRET=$CONFLUENT_API_SECRET
```



The wrapper script (scripts/deploy-cc-flink-reports.sh) builds the PTF/UDAF shadow JAR if missing, then runs `terraform apply -auto-approve` in terraform/. First-run takes ~6–8 minutes (Kafka cluster provisioning dominates). Re-applies are idempotent — `CREATE … IF NOT EXISTS` plus `lifecycle { ignore_changes = [compute_pool] }` on every statement.

What gets created (see terraform/setup-confluent-flink.tf for the full graph):

Resource	Name	Notes
`confluent_environment`	`confluent-kafka-isotope`	ESSENTIALS stream-governance package
`confluent_kafka_cluster`	`kafka-isotope`	Standard, single-zone, AWS us-east-1 by default

Resource	Name	Notes
`confluent_kafka_topic` × 4	`orders.{placed,enriched,fulfilled}` + `platform.observability.consume_events`	Only the event topics + the consume-marker topic are pre-created. The 6 `isotope_report_*_1m` sink topics are created on first deploy by their `CREATE TABLE` statement (CCAF's Topic Catalog auto-imports any pre-existing topic as `(key BYTES, val BYTES)`, which would silently no-op the typed `CREATE TABLE`). `terraform destroy` cleans them up via the environment cascade.
`confluent_service_account` + 6 role bindings	`isotope-flink-sql-runner`	FlinkDeveloper (org) + ResourceOwner on topic=* / transactional-id=* / group=* / SR subject=* + Assigner on the service account
`confluent_flink_compute_pool`	`isotope-flink-statement-runner`	10 CFU; headroom for 6 INSERTs + ad-hoc SELECTs
`confluent_flink_artifact`	`isotope-flink-udf`	Uploads `ptf/build/libs/isotope-flink-udf.jar`
`confluent_flink_statement` × 20	(see file)	4 ALTER TABLE (event-topic scalar headers) + 3 VIEW (raw + typed produce + typed consume) + 6 sink CREATE TABLE + 1 CREATE FUNCTION (PTF only; UDAF skipped) + 6 INSERT INTO

Useful outputs:

```
terraform -chdir=terraform output environment_id
terraform -chdir=terraform output kafka_bootstrap_servers
terraform -chdir=terraform output schema_registry_url
terraform -chdir=terraform output -raw kafka_api_key     # sensitive
terraform -chdir=terraform output -raw kafka_api_secret  # sensitive
```

Format-by-runtime (not -by-domain). CP's reports land on **Avro+SR** (`'value.format' = 'avro-confluent'` in scripts/flink/sql/cp/05_report_sinks.fql). CCAF's reports land on **Protobuf+SR** (`'value.format' = 'proto-registry'` in each sink's WITH clause in terraform/setup-confluent-flink.tf). The two runtimes' SQL is otherwise unshared: CP's lives hardcoded in scripts/flink/sql/cp/, CCAF's lives inline as `confluent_flink_statement` resources in terraform/setup-confluent-flink.tf.

CCAF UDAF limitation. CCAF currently rejects all `CREATE FUNCTION` statements for user-defined aggregate functions ("aggregate functions are not supported"). The `LATENCY_PERCENTILES` UDAF therefore deploys on CP only; the percentile report (`latency_percentiles_flat_1m`) does not exist on CCAF. The other JAR-backed function, `STUCK_TRACE_PTF` (a ProcessTableFunction, not a UDAF), works on both runtimes. The JAR itself is portable — `LatencyPercentilesUDAF` ships with a byte[]-based accumulator and is ready to register when Confluent lifts the restriction. The six CCAF reports are: `latency` (avg/min/max), `topology` (produce-side), `bipartite_topology` (full service↔topic↔service graph), `hop_distribution`, `coverage`, `stuck_trace`.

Driving traffic — the 4-stage demo against CCAF. App.java reads four optional `-D` properties (`kafka.security.protocol`, `kafka.sasl.mechanism`, `kafka.sasl.jaas.config`, `schema.registry.basic.auth.user.info`) that default to plaintext-no-auth for Minikube. scripts/cc-cli-env.sh pulls the Kafka + Schema-Registry credentials from `terraform output` (both keys are rotated by `module.kafka_api_key_rotation` and `module.sr_api_key_rotation` in terraform/setup-confluent-kafka.tf) and builds the JAAS string.

The thin wrapper scripts/cc-app-run.sh sources the env helper then invokes `./gradlew :app:run` with the six `-D` flags. It accepts two argument styles: **pipeline-position verbs** (`place` / `enrich` / `fulfill` / `ship`) that encode the orders.* topic chain so the 4-terminal demo is one word per terminal, and the **generic `send` / `hop` / `consume` / `sink` passthrough** for ad-hoc inspection on topics outside the demo. Run the script with no args for the full verb list.

```
# Four terminals (same A/B/C/D order as § 4.2). No manual env exports —
# the wrapper sources cc-cli-env.sh, which pulls everything from terraform.
scripts/cc-app-run.sh ship              # A — terminal consumer (emits marker)
scripts/cc-app-run.sh fulfill           # B
scripts/cc-app-run.sh enrich            # C
scripts/cc-app-run.sh place 'hello'     # D
```

The wrapper hard-fails with a clear message if any of the seven required values is missing, so you'll never silently hand gradle empty `-D` values.

Terminal A prints the same `trace_id` across all three hops, and the CCAF report INSERTs populate as you fire Terminal D — `SELECT * FROM isotope_report_latency_1m`, `SELECT * FROM isotope_report_bipartite_topology_1m`, etc. in the Cloud Console SQL workspace. The bipartite report shows all six edges of the chain (3 produce + 3 consume).

Sustained traffic — required to see report rows. The six INSERT INTO jobs aggregate over `TUMBLE(event_time, INTERVAL '1' MINUTE)` windows, and a tumbling window only emits when the watermark advances past `window_end`. A handful of records bursted from Terminal D within a single 1-minute interval will sit in one open window forever (the most-recent record is the watermark, and it never gets older than itself). Spread traffic across **multiple** windows so the watermark crosses each boundary:

```
# 30 records spaced 5s apart ≈ 2.5 minutes of event-time → spans 3+ windows
for i in {1..30}; do
  scripts/cc-app-run.sh place "burst-$i"
  sleep 5
done
```

Wait ~90 seconds after the *last* record before checking `isotope_report_latency_1m` (and friends) — that's the watermark catching up. The `stuck_trace_alerts_1m` sink only fires for traces that go ≥60s of event time without a fresh hop, so the burst above won't trigger it (every trace gets one record and ends — no stalled in-flight state). To exercise `STUCK_TRACE_PTF`: send a single record to `orders.placed` and don't run the `order-enrichment-service` / `order-fulfillment-service` hops, then keep sending unrelated records elsewhere so the watermark advances past the stuck trace's `event_time + 60s`.

Teardown:

```
make cc-flink-reports-down CONFLUENT_API_KEY=$CONFLUENT_API_KEY
CONFLUENT_API_SECRET=$CONFLUENT_API_SECRET
```

Runs `terraform destroy -auto-approve` — deletes every resource above, including the environment itself. Safe to run repeatedly.